

Diagnostic report -- data severe drift

Overall status: CRITICAL Generated at 2026-08-30T18:04:08.599178+00:00 | Cycle (most recent signal): 2026-08-30T18:04:08.587672+00:00

Executive summary

Data drift: 5/8 feature(s) drifted (severity critical). Target drift (proxy): PSI=0.000, severity warning. Alerts: 1 critical, 2 warning.

1. Data / Feature Drift

Overall severity: **critical** -- 5/8 feature(s) drifted (62.5%).

Feature	Type	PSI	Wasserstein (norm)	Severity	Drifted	Quality
Pregnancies	numeric	0.0090	0.0174	none	no	ok
Glucose	numeric	2.0510	1.5349	critical	yes	ok
BloodPressure	numeric	3.7542	0.8923	critical	yes	ok
SkinThickness	numeric	0.0339	6.9195	warning	yes	ok
Insulin	numeric	10.6260	2.3314	critical	yes	29.9% missing
BMI	numeric	1.0948	0.9875	critical	yes	ok
DiabetesPedigreeFunction	numeric	0.0074	0.0273	none	no	ok
Age	numeric	0.0122	0.0200	none	no	ok

2. Target Drift

Status: **proxy** | n_samples = 768

PSI = 0.0000 | severity = **warning** | drifted = yes Wasserstein distance (normalized) = 0.2396 Reference mean = 0.3490 -> current mean = 0.5885 (reference std = 0.4766 -> current std = 0.4921) *Note: Proxy signal (predictions), needs confirmation with true labels.*

Alerts raised this cycle (3)

[CRITICAL] High model-impact drift on 3 feature(s) (source: data_drift)

Affected features: Glucose, SkinThickness, Insulin. - Glucose: 'Glucose' is at the 88th percentile of SHAP importance among tracked features (threshold=75th, raw importance=0.5135): the drift directly affects predictions. - SkinThickness: 'SkinThickness' is at the 75th percentile of SHAP importance among tracked features (threshold=75th, raw importance=0.2962): the drift directly affects predictions. - Insulin: 'Insulin' is at the 100th percentile of SHAP importance among tracked features (threshold=75th, raw importance=1.5079): the drift directly affects predictions.

Recommended action (expert): FINAL VERDICT -- pipeline issue confirmed, do NOT retrain yet: Insulin: 29.9% missing values this batch. Fix ingestion for Insulin first (schema, upstream source, mapping of new categories), then re-run this cycle on clean data before deciding whether real drift remains.

Remediation type: data_pipeline_investigation

[WARNING] Drift to monitor on 2 feature(s) (source: data_drift)

Affected features: BloodPressure, BMI. - BloodPressure: 'BloodPressure' has low SHAP importance (percentile 25) and isn't correlated with an influential feature: likely negligible model impact despite the statistical drift. - BMI: 'BMI' has low SHAP importance (percentile 50) and isn't correlated with an influential feature: likely negligible model impact despite the statistical drift.

Recommended action (expert): No immediate action: risk is indirect (correlated with a key feature) or undetermined (missing SHAP/correlation data). Re-evaluate next cycle.

Remediation type: monitor

[WARNING] Early target-drift signal (proxy, unconfirmed) (source: target_drift)

PSI on predictions (0.000) exceeds the drift threshold. True labels not yet available: needs confirmation.

Recommended action (expert): Don't retrain on this signal alone -- it's drift on PREDICTIONS, not ground truth. Wait for the next labeled cycle to confirm; investigate manually if it persists unconfirmed for 2-3 cycles.

Remediation type: monitor

Remediation actions (3)

Type	Urgency (SLA)	Status	Approval required	Automatable	Source alert
data_pipeline_investigation	urgent (within 24h)	executed	no	yes	High model-impact drift on 3 feature(s)

monitor	normal (next working cycle)	executed	no	yes	Drift to monitor on 2 feature(s)
monitor	normal (next working cycle)	executed	no	yes	Early target-drift signal (proxy, unconfirmed)

Generated automatically by the drift monitoring pipeline.